IT5041
Full Stack Software Development with Python

Assignment 2
Flask Web App with MongoDB Storage and Retrieval

Submitted by

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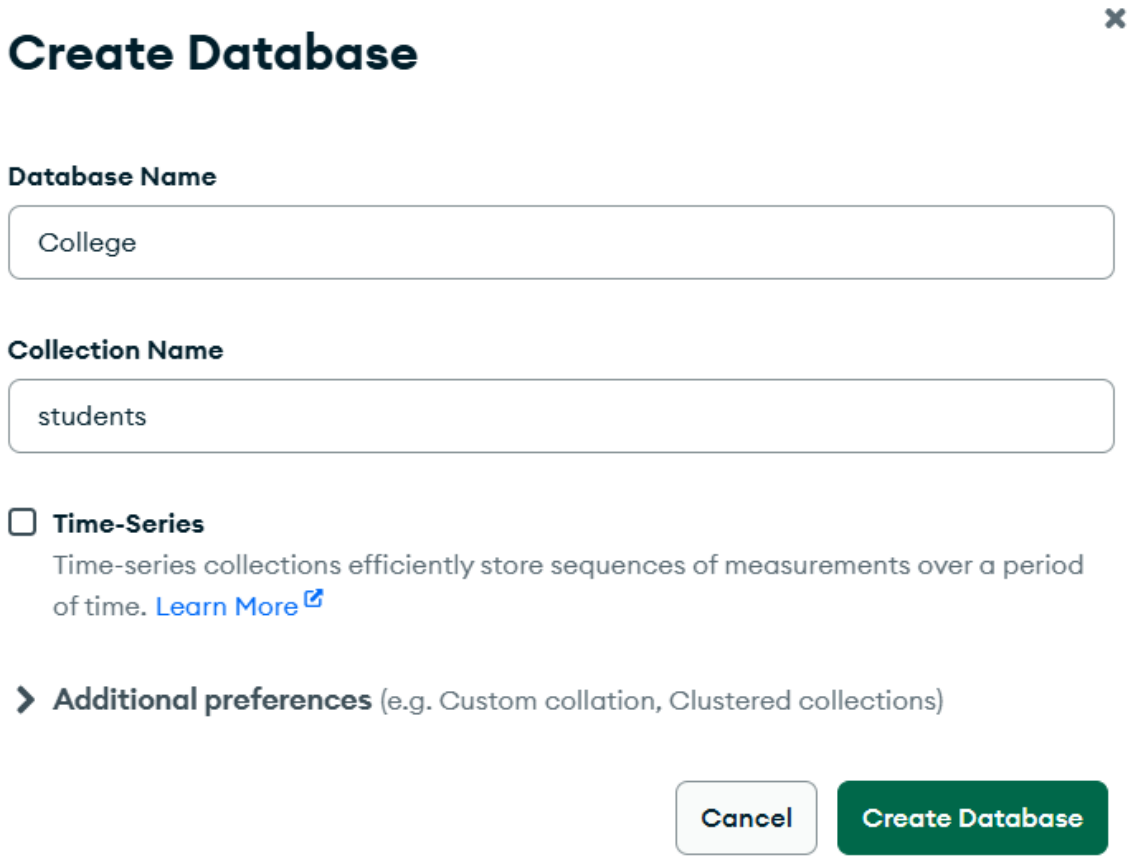
Database and Collection in MongoDB Compass



add_student.html

```
<!DOCTYPE html>
<html lang="en">
<head>
   <meta charset="UTF-8">
   <title>Add Student</title>
   <link rel="stylesheet" href="{{ url_for('static',
filename='add_student.css') }}">
   <script>
      function addSubjectField() {
         const container = document.getElementById('subjects-container');
         const input = document.createElement('input');
         input.type = 'text';
         input.name = 'subjects[]';
         input.placeholder = 'Enter Subject';
         container.appendChild(input);
      }
```

```
    </script>
</head>
<body>
    <header>
        <h1>College Management System</h1>
        <nav>
            <a href="{{ url_for('add_student') }}">Add Student</a>
            <a href="{{ url_for('view_students') }}">View Student Details</a>
        </nav>
    </header>

    <div class="container">
        <h2>Add Student</h2>
        <form method="POST" action="{{ url_for('save_student') }}">
            <label>Register Number:</label>
            <input type="text" name="register_number" required>

            <label>First Name:</label>
            <input type="text" name="first_name" required>

            <label>Last Name:</label>
            <input type="text" name="last_name" required>

            <label>Department:</label>
            <select name="department" required>
                {% for dept in departments %}
                    <option value="{{ dept }}">{{ dept }}</option>
                {% endfor %}
            </select>

            <label>Semester:</label>
            <input type="number" name="semester" min="1" max="8" required>

            <label>Subjects:</label>
            <div id="subjects-container">
                <input type="text" name="subjects[]" placeholder="Enter Subject">
            </div>
            <button type="button" onclick="addSubjectField()">Add Another
Subject</button>
```

```
            <button type="submit">Save Student</button>
        </form>
    </div>
</body>
</html>
```

add_student.css

```
body {
    font-family: Arial, sans-serif;
    background: #f4f6f8;
    margin: 0;
}

header {
    background: #003366;
    color: white;
    padding: 10px 20px;
}

header h1 {
    margin: 0;
}

nav {
    margin-top: 10px;
}

nav a {
    color: white;
    margin-right: 20px;
    text-decoration: none;
    font-weight: bold;
}

.container {
    background: white;
    width: 50%;
    margin: 30px auto;
    padding: 20px;
    border-radius: 8px;
```

```
    box-shadow: 0px 0px 8px #ccc;
}

label {
    display: block;
    margin-top: 10px;
}

input, select {
    width: 95%;
    padding: 8px;
    margin-top: 5px;
}

button {
    background: #003366;
    color: white;
    padding: 10px 15px;
    border: none;
    margin-top: 15px;
    cursor: pointer;
    border-radius: 4px;
}

button:hover {
    background: #0055aa;
}
```

view_students.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>View Students</title>
    <link rel="stylesheet" href="{{ url_for('static',
filename='view_students.css') }}">
</head>
<body>
    <header>
        <h1>College Management System</h1>
```

```
    <nav>
        <a href="{{ url_for('add_student') }}">Add Student</a>
        <a href="{{ url_for('view_students') }}">View Student Details</a>
    </nav>
</header>

<div class="container">
    <h2>Student Details</h2>
    {% if students %}
    <table>
        <thead>
            <tr>
                <th>Register Number</th>
                <th>First Name</th>
                <th>Last Name</th>
                <th>Department</th>
                <th>Semester</th>
                <th>Subjects</th>
            </tr>
        </thead>
        <tbody>
            {% for student in students %}
            <tr>
                <td>{{ student['Register Number'] }}</td>
                <td>{{ student['First Name'] }}</td>
                <td>{{ student['Last Name'] }}</td>
                <td>{{ student['Department'] }}</td>
                <td>{{ student['Semester'] }}</td>
                <td>
                    <ul>
                        {% for subject in student['Subjects'] %}
                            <li>{{ subject }}</li>
                        {% endfor %}
                    </ul>
                </td>
            </tr>
            {% endfor %}
        </tbody>
    </table>
    {% else %}
```

```
        <p>No students found.</p>
        {% endif %}
    </div>
</body>
</html>
```

view_students.css

```
body {
    font-family: Arial, sans-serif;
    background: #f4f6f8;
    margin: 0;
}

header {
    background: #003366;
    color: white;
    padding: 10px 20px;
}

header h1 {
    margin: 0;
}

nav a {
    color: white;
    margin-right: 20px;
    text-decoration: none;
    font-weight: bold;
}

.container {
    background: white;
    width: 90%;
    margin: 30px auto;
    padding: 20px;
    border-radius: 8px;
    box-shadow: 0px 0px 8px #ccc;
}

table {
```

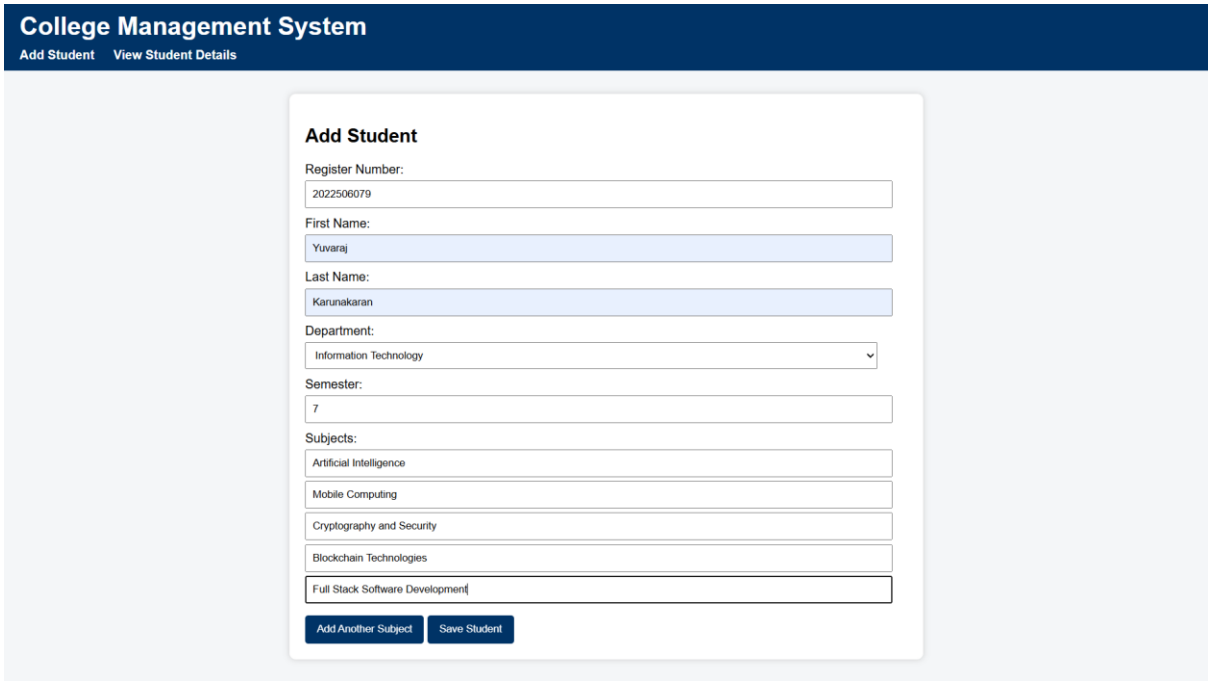
```
    width: 100%;
    border-collapse: collapse;
    margin-top: 15px;
}

th, td {
    border: 1px solid #ccc;
    padding: 10px;
    text-align: left;
}

th {
    background-color: #003366;
    color: white;
}

ul {
    margin: 0;
    padding-left: 20px;
}
```

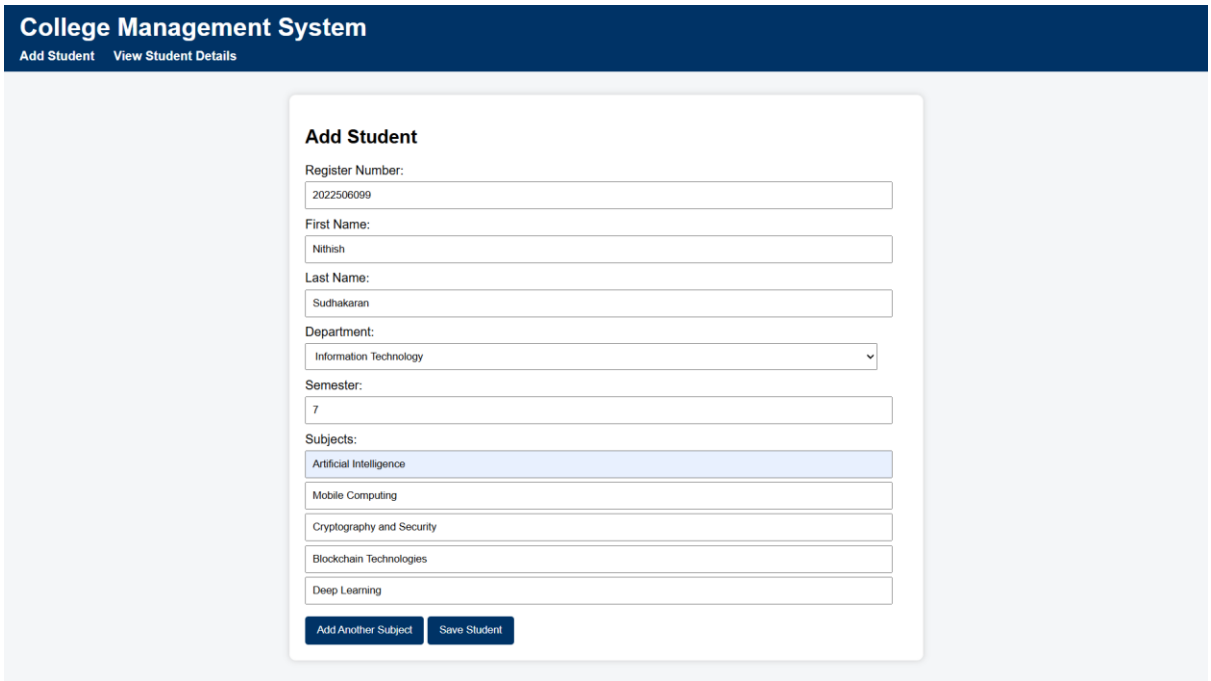
app.py

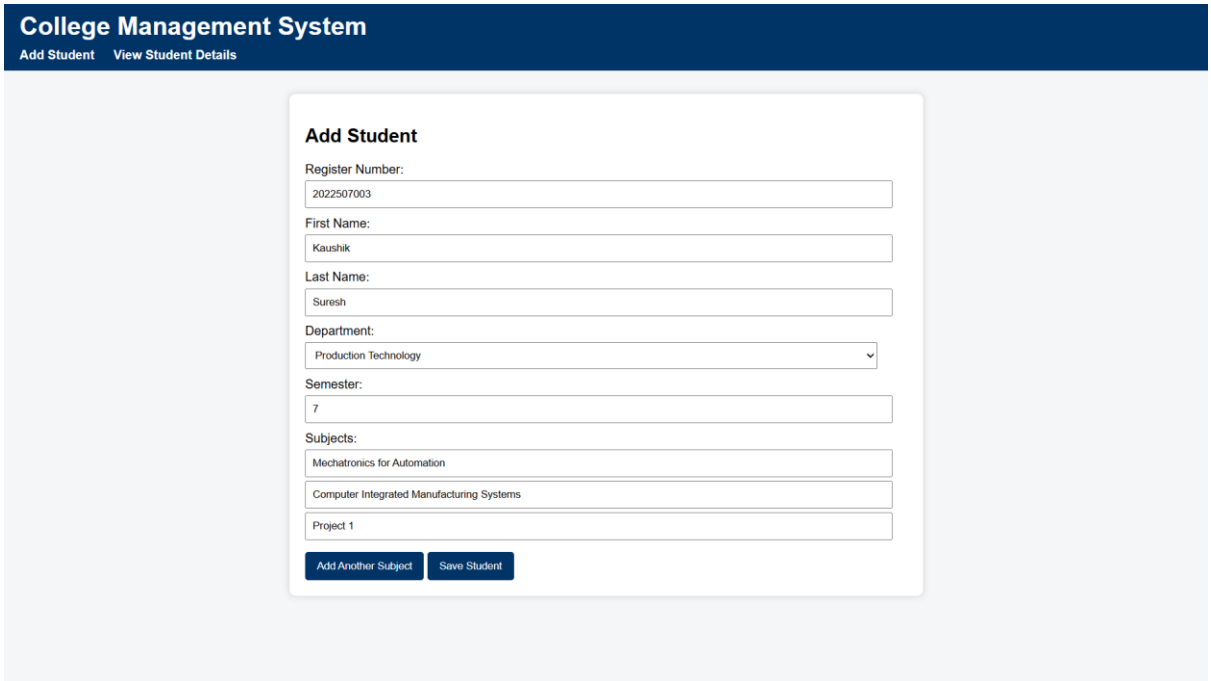
```
from flask import Flask, render_template, request, redirect, url_for
from pymongo import MongoClient

app = Flask(__name__)

# MongoDB connection
client = MongoClient("mongodb://localhost:27017/")
db = client["College"]
collection = db["students"]

@app.route('/')
def add_student():
    departments = [
        "Aeronautical Engineering",
        "Automobile Engineering",
        "Electronics Engineering",
        "Instrumentation Engineering",
        "Production Technology",
```

```
        "Rubber and Plastics Technology",
        "Information Technology",
        "Computer Technology",
        "Applied Sciences and Humanities"
    ]
    return render_template("add_student.html", departments=departments)

@app.route('/add', methods=['POST'])
def save_student():
    reg_no = request.form.get("register_number")
    first_name = request.form.get("first_name")
    last_name = request.form.get("last_name")
    department = request.form.get("department")
    semester = request.form.get("semester")
    subjects = request.form.getlist("subjects[]")

    data = {
        "Register Number": reg_no,
        "First Name": first_name,
        "Last Name": last_name,
        "Department": department,
        "Semester": semester,
        "Subjects": subjects
    }

    collection.insert_one(data)
    return redirect(url_for('view_students'))

@app.route('/view')
def view_students():
    students = list(collection.find({}, {'_id': 0}))
    return render_template("view_students.html", students=students)

if __name__ == "__main__":
    app.run(debug=True)
```

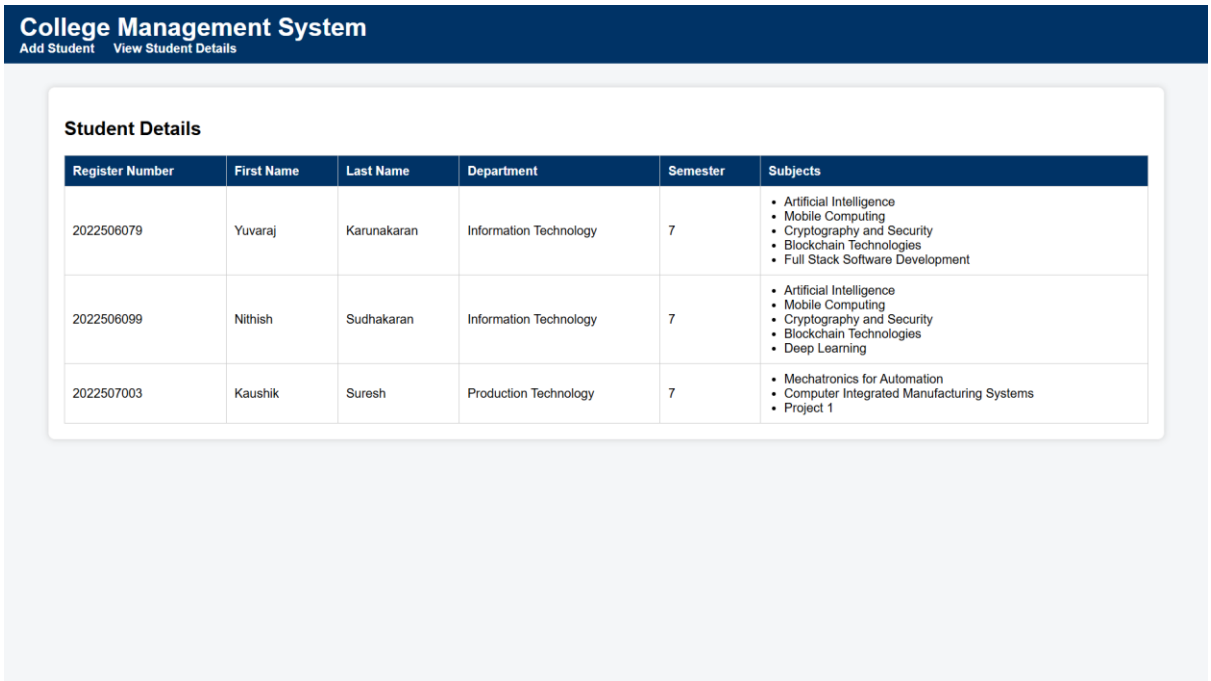
Adding New Students





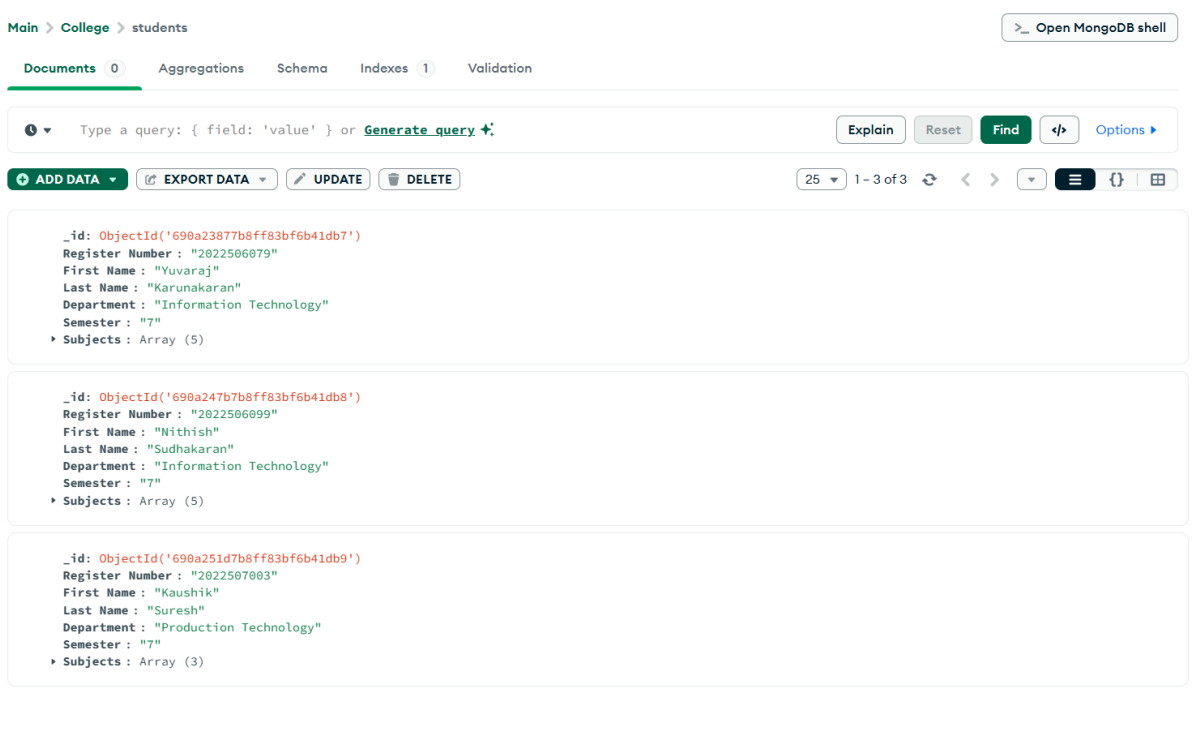


Viewing Student Details



Register Number	First Name	Last Name	Department	Semester	Subjects
2022506079	Yuvaraj	Karunakaran	Information Technology	7	• Artificial Intelligence • Mobile Computing • Cryptography and Security • Blockchain Technologies • Full Stack Software Development
2022506099	Nithish	Sudhakaran	Information Technology	7	• Artificial Intelligence • Mobile Computing • Cryptography and Security • Blockchain Technologies • Deep Learning
2022507003	Kaushik	Suresh	Production Technology	7	• Mechatronics for Automation • Computer Integrated Manufacturing Systems • Project 1

Database – MongoDB Compass



NoSQL Data as JSON


```
[{
  "_id": {
    "$oid": "690a23877b8ff83bf6b41db7"
  },
  "Register Number": "2022506079",
  "First Name": "Yuvaraj",
  "Last Name": "Karunakaran",
  "Department": "Information Technology",
  "Semester": "7",
  "Subjects": [
    "Artificial Intelligence",
    "Mobile Computing",
    "Cryptography and Security",
    "Blockchain Technologies",
    "Full Stack Software Development"
  ]
},
{
  "_id": {
    "$oid": "690a247b7b8ff83bf6b41db8"
```

```
  },
  "Register Number": "2022506099",
  "First Name": "Nithish",
  "Last Name": "Sudhakaran",
  "Department": "Information Technology",
  "Semester": "7",
  "Subjects": [
    "Artificial Intelligence",
    "Mobile Computing",
    "Cryptography and Security",
    "Blockchain Technologies",
    "Deep Learning"
  ]
},
{
  "_id": {
    "$oid": "690a251d7b8ff83bf6b41db9"
  },
  "Register Number": "2022507003",
  "First Name": "Kaushik",
  "Last Name": "Suresh",
  "Department": "Production Technology",
  "Semester": "7",
  "Subjects": [
    "Mechatronics for Automation",
    "Computer Integrated Manufacturing Systems",
    "Project 1"
  ]
}]
```